

Haadia Farrukh

Email: haadia.farrukh.work@gmail.com LinkedIn: <https://www.linkedin.com/in/h-farrukh> Phone : (+92)3092194816

SUMMARY

Embedded Controls Engineer with **2+** years delivering **real-time** firmware for hybrid solar inverters on **TI C2000** DSPs. Specialized in **high-speed control loops**, **grid synchronization**, and **embedded safety systems**. Proven in **optimizing ISR timing**, **boosting system reliability**, and shipping hardware-ready firmware from concept to field deployment.

WORK EXPERIENCE

CARE Pvt. Ltd. – Center for Advanced Research in Engineering – Islamabad, Pakistan

Sr Design Engineer – 07/2025 – Current

General

- Diagnosed and resolved **30+ critical field failures** in firmware and hardware, delivering fixes that **boosted MTBF by 20%**.
- Led **bring-up and validation** of **5+ custom PCB assemblies** (power, control, and communication), identifying and correcting flaws that prevented costly re-spins.
- **Trained and mentored** 3 interns from Simulink simulation through DSP deployment and hardware debugging;
- Conducted **15+ technical interviews** for embedded and power electronics roles, making final recommendations that built the R&D team's talent base.

Design Engineer – 07/2023 – 07/2025

Project : 3KW Hybrid Solar Inverter

- Modeled and verified **PLL, MPPT, battery buck-boost control**, and **PR-based grid voltage/current regulation** in MATLAB/Simulink with realistic non-idealities (grid harmonics, switching losses, battery current spikes).
- Ported the complete control stack to **TI C2000 F2800137** — the smallest, lowest-cost DSP in its family — optimizing ISR execution time to **50 μ s** to meet a strict **control parameters update window**.
- Designed an **event-driven state machine** for seamless source prioritization (grid, PV, battery) with embedded safety-first logic.
- Configured **ePWM** (active-high complementary, dead-time insertion), precision-timed **ADC sampling** at current waveform midpoints, and **SPI** interface for cloud telemetry.
- Implemented comprehensive **protection systems** (over/undervoltage, overcurrent, thermal shutdown, anti-islanding, source loss detection), preventing catastrophic failures.
- Delivered a **cost-competitive design** that ran advanced multi-loop control on a budget DSP, preserving BOM costs while achieving grid-tied $\pm 1\%$ voltage regulation and stable standalone operation.

Project : SDR : Software Defined Radio

- Implemented and optimized narrowband (NB) signal SNR estimation algorithms on FPGA (Xilinx), DSP (TI), and ARM Cortex-M platforms, improving detection accuracy by 15% in noisy channels.
- Authored a publicly available SDR knowledge base, covering beginner-to-advanced FPGA/DSP workflows, which attracted 100+ visits and improved team onboarding efficiency.

xFlow Research (Pvt.) Ltd. – Islamabad, Pakistan

Intern – 06/2022 – 08/2022

- Conducted comprehensive research on various **codecs** to develop a codec analysis program, and collaborated with the Data Analytics team to effectively analyze audio data.
- Deployed **Devstack** and learned the basics of **Cloud architecture**
- Learned to proficiently create and manage **virtual machines** with Linux operating systems, while gaining expertise in **configuring** and **troubleshooting** computer networks to ensure seamless communication

EDUCATION

Bachelor of Electrical Engineering NUST National University of Science and Technology

2019 – 2023 Islamabad, Pakistan

Elective Coursework:

Digital System Design, Computer Communication Networks, Embedded System Design

DPDK-Based Acceleration of DNS for Cloud Environments

Final Year Project done in collaboration with xFlow Research (Pvt.) Ltd

- Engineered a high-performance DNS server solution by deploying Bind9 and conducting latency, packet loss, and throughput benchmarking.
- Integrated DPDK (Data Plane Development Kit) for kernel bypass, developing custom C++ networking modules to improve packet processing speed by 45%.
- Implemented a full TCP/IP stack in user space, reducing average DNS query latency from 2.3 ms to 1.2 ms.
- Identified and resolved network bottlenecks, improving reliability under 10,000+ concurrent client load.

Real-Time Fire Alarm Detection with Wi-Fi Alerts

Course: Embedded System Design

- Designed and developed a fire alarm detection system using an ESP32 microcontroller and an INMP441 microphone, capable of accurately processing audio and sending instant notifications via webpage and email, ensuring timely alerts to authorities in the event of a fire alarm detection.

SKILLS

Programming & Firmware : C, Embedded C, C++, Python, MATLAB/Simulink, FreeRTOS

Hardware: TI C2000 DSPs, STM32F4 series, ARM Cortex-M, FPGAs (Altera/Intel), 8051, 8086 , Arduino, ESP-32

Control & Power Electronics: PID tuning, MPPT algorithms, PWM generation, grid-tied inverter control,

Networking & High-Performance Computing: DPDK, TCP/IP stack, Bind9 DNS, kernel bypass optimization

Tools & Software: Altera Quartus II, Proteus, CCS, Linux, JTAG/ISP programming, oscilloscopes, logic analyzers